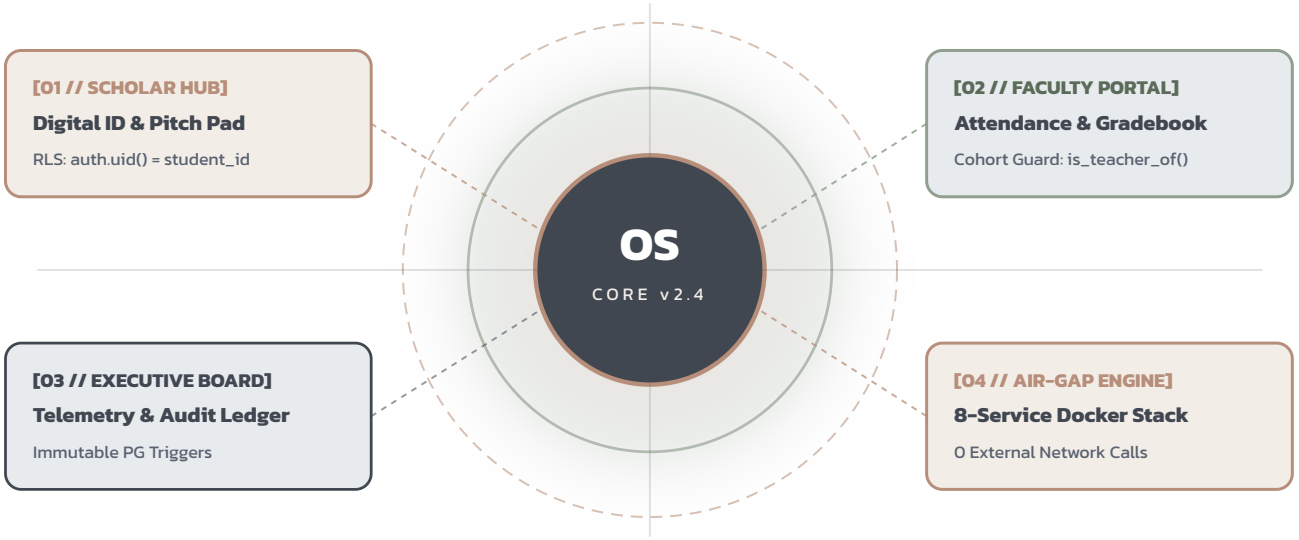
• AIR-GAPPED SOVEREIGN STACK | DPDP ACT 2023 COMPLIANT | ZERO CLOUD TELEMETRY

THE COGNITIVE *Institutional* OPERATING SYSTEM.

An end-to-end, self-hosted, zero-trust school operating system uniting **Scholars**, **Educators**, and **Executive Leadership** on a single immutable, air-gapped local canvas.



0 B	24 / 24	8 Nodes	< 14ms
EXTERNAL CLOUD LEAKAGE	RLS-PROTECTED TABLES	ON-PREM DOCKER TOPOLOGY	LOCAL LAN SYNC LATENCY

EXECUTIVE OVERVIEW

Why We Built a Sovereign Operating System for Schools

Modern educational institutions are trapped between archaic paper registers and predatory multi-tenant cloud SaaS platforms that harvest minor student data. **Campusly / Eduvia (BrainzOS)** resolves this dichotomy by delivering a consumer-grade, real-time glassmorphic web workspace backed by an air-gapped, self-hosted 8-service PostgreSQL & Supabase infrastructure.

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"Education without friction is not just a dream — it is a sovereign, self-hosted operating system where every student record remains strictly inside the school walls."

THE STATUS QUO CRISIS

Four Systemic Failures Paralyzing Modern Schools

Across 1.5M+ schools in India and globally, administrators and educators lose up to **34% of instructional hours** navigating disconnected spreadsheets, WhatsApp groups, and insecure third-party portals.

FAILURE MODE // 01

Fragmented Multi-Tool Chaos

Attendance lives on paper registers, marks in Excel files, homework on WhatsApp, timetables on PDF notices, and leave approvals on paper slips. Zero real-time correlation exists between student attendance drops and academic performance.

4–7 Disconnected Tools Used Daily Per Teacher

FAILURE MODE // 02

Cloud Data Harvesting of Minors

Conventional cloud ERPs store sensitive minor PII (biometrics, medical history, disciplinary records, parental contacts) on shared third-party SaaS databases outside the school's physical control — violating strict DPDP Act 2023 principles.

100% Third-Party Exposure in Standard SaaS ERPs

FAILURE MODE // 03

Predatory Per-Student SaaS Rent

Cloud vendors charge recurring per-student/per-month fees and lock historical institutional transcripts behind proprietary paywalls. If a school stops paying, it loses years of operational continuity.

₹8L–₹25L ($10k–$30k) Annual Recurring Drain

FAILURE MODE // 04

Internet Outage Paralysis & Hardcoded Rigidity

When campus broadband drops, cloud ERPs freeze completely during morning roll-call. Worse, legacy ERPs hardcode specific board structures, breaking when schools adopt NEP 2020, IB, or Cambridge modular streams.

Zero Offline Resilience in Legacy Web Portals

COMPARATIVE BENCHMARK: LEGACY CLOUD ERP VS. MANUAL REGISTERS VS. CAMPUSLY / EDUVIA

OPERATIONAL DIMENSION	PAPER & EXCEL REGISTERS	LEGACY CLOUD SAAS ERP	CAMPUSLY / EDUVIA (OUR SOLUTION)
Student Data Residency	Physical Lockers (Fragile)	Third-Party Cloud Servers	**100% On-Premise School Hardware**
Works Without Internet	Yes (No Analytics)	No (Total Outage)	**Yes (Air-Gapped LAN at <14ms)**
Row-Level Security (RLS)	None	App-Layer Only (IDOR Prone)	**PostgreSQL Engine Enforced (24 Tables)**
Curriculum Adaptability	Manual Re-entry	Vendor Ticket Required	**Instant CBSE / ICSE / IB / IGCSE Presets**

ARCHITECTURAL BREAKTHROUGH

Every Student. Every Teacher. Every Director. One Sovereign Canvas.

Instead of maintaining three separate web apps that drift out of sync, our platform consolidates all **30+ operational views** into a unified, zero-build institutional shell with instantaneous Server-Sent Events (SSE) & WebSocket synchronization.

ROLE 01 // SCHOLAR

Student Experiential Hub

Empowering agency, transparency, and entrepreneurship.

- **Live Telemetry Dashboard:** Real-time attendance %, house rank, and upcoming exam countdowns.
- **Digital ID Pass:** Cryptographically tied campus identity card with stream & cohort badges.
- **Continuous Assessment Ledger:** Instant visibility into unit test, mid-term, and final grades.
- **Pitch Pad Incubator:** Student startup showcase with tamper-proof peer voting triggers.
- **Coursework & Broadcasts:** Direct assignment submission & school-wide dispatches.

ROLE 02 // EDUCATOR

Faculty Command Center

Eliminating paperwork so teachers can focus on teaching.

- **1-Click Period Attendance:** Rapid cohort roll-call with grouped attendance history.
- **Dynamic Gradebook:** Weighted continuous evaluation & final exam marks entry.
- **Lecture & Test Planner:** Syllabus pacing, lesson plans, and examination scheduling.
- **Class Record Register:** Comprehensive 360° student dossiers & document attachments.
- **Leave & Broadcasts:** Instant leave applications and targeted class announcements.

ROLE 03 // EXECUTIVE

Director & Principal Board

Total institutional command and forensic accountability.

- **Executive Telemetry Board:** School-wide attendance rollups & cohort analytics.
- **Zero-Code Academic Setup:** 1-click board presets (CBSE, ICSE, IB, IGCSE, State) + custom CRUD.
- **Faculty & Student Provisioning:** Bulk CSV student import via `SECURITY DEFINER` RPC.
- **Leave & Substitutions:** Approve faculty leave and assign period substitutes in seconds.
- **Immutable Audit Ledger:** Append-only forensic log of every sensitive database mutation.

END-TO-END SYNCHRONIZATION PIPELINE (REAL-TIME CROSS-ROLE FLOW)

STEP 01

Teacher Action

Marks Period Attendance or publishes Unit Test Grades in Class XI-B.

→

STEP 02

PostgreSQL + RLS

Verifies `is_teacher_of()`, commits transaction & writes Audit Log.

→

STEP 03

Realtime SSE Bus

Elixir WebSocket / SSE broadcasts delta across local LAN in <14ms.

→

STEP 04

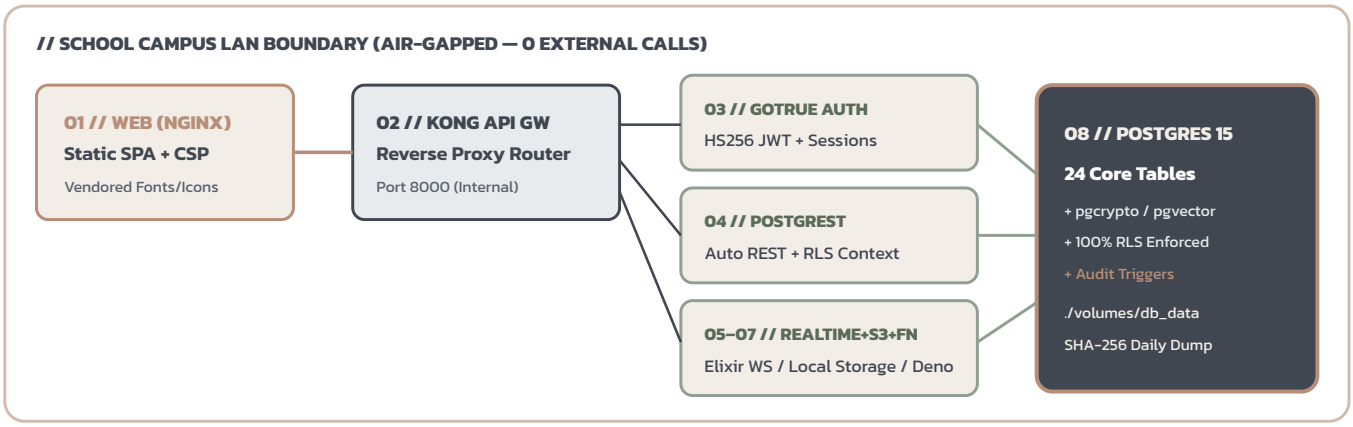
Student & Director

Student dashboard GPA/Attendance & Director Rollup update live without reload.

SOVEREIGN INFRASTRUCTURE

Self-Hosted 8-Container Topology (`docker-compose.yml`)

Designed to run on a single school server (Linux or Windows Server) over the campus LAN/Wi-Fi. All 8 microservices communicate over an isolated internal Docker bridge network, exposing **only Port 80/443** to campus devices.



CONTAINER SERVICE SPECIFICATION

1. db (PostgreSQL 15)	pgcrypto, pgvector, 24 tables, WAL
2. kong (API Gateway)	Unified routing (/auth, /rest, /storage)
3. auth (GoTrue)	JWT issuance, password policy, allowlist
4. rest (PostgREST)	Zero-boilerplate SQL-to-REST with RLS
5. realtime (Elixir)	Instant cross-tab & cross-role WebSocket
6. storage (Local S3)	Encrypted local disk bucket for dossiers
7. functions (Deno)	Isolated server-side business logic
8. web (Nginx + Node)	Hardened HTTP headers + local vendor assets

DUAL-MODE RUNTIME: DOCKER OR LIGHTWEIGHT NODE

In addition to the full 8-container Docker stack, we engineered `scripts/serve.js` — a zero-dependency Node.js local server implementing a complete Supabase-compatible API (`/auth/v1`, `/rest/v1`, `/realtime/v1/sse`) on the same origin for instant exhibition demos and low-resource school hardware.

```
# Boot Full Production Air-Gap Stack
$ node scripts/generate-keys.js
$ docker compose up -d
$ node scripts/bootstrap-director.js --email director@school.edu
```

DEFENSE-IN-DEPTH ENGINEERING

20-Point Institutional Security Hygiene Matrix

Every layer of the stack — from Git commit hooks to browser Content Security Policy (CSP) headers and PostgreSQL row-level policies — is engineered on a **Fail-Closed, Zero-Trust** security posture.

10 CORE CRYPTOGRAPHIC & ACCESS CONTROLS

01. 100% Row-Level Security	Enforced on all 24 PostgreSQL tables
02. Zero Service-Role Exposure	Browser receives only restricted ANON_KEY
03. IDOR Prevention	auth.uid() ownership verified on every row
04. Pre-Commit Secret Guard	.gitleaks.toml blocks JWTs & keys in Git
05. Fail-Closed Role Routing	navigateTo() blocks unauthorized URL hashes
06. Role Escalation Trigger	DB trigger blocks non-director role edits
07. Ephemeral Key Injection	crypto.randomBytes(64) via generate-keys.js
08. Forced First-Login Reset	must_change_password flag on new accounts
09. XSS Output Encoding	Strict escapeHtml() & textContent rendering
10. Domain Allowlist Gate	DB-backed school email domain enforcement

HARDENED HTTP SECURITY HEADERS (NGINX / VERCEL / NETLIFY)

```
X-Frame-Options: DENY
X-Content-Type-Options: nosniff
Referrer-Policy: no-referrer
Permissions-Policy: camera=(), microphone=(), geolocation=(), payment=()
Strict-Transport-Security: max-age=63072000; includeSubDomains; preload
Content-Security-Policy: default-src 'self'; object-src 'none'; frame-ancestors 'none';
```

CRYPTOGRAPHIC KEY ROTATION PROTOCOL

If a school IT administrator ever suspects credential compromise, running `node scripts/generate-keys.js` regenerates a 32-byte DB password, 64-byte hex JWT secret, and newly signed HS256 `anon` and `service_role` tokens — invalidating all active sessions across campus in under 5 seconds.

DATABASE-LEVEL PRIVILEGE ESCALATION DEFENSE (SQL TRIGGER VERIFICATION)

```
CREATE OR REPLACE FUNCTION public.prevent_role_escalation() RETURNS TRIGGER AS $$
BEGIN
  IF NEW.role IS DISTINCT FROM OLD.role AND NOT public.is_director() THEN
    RAISE EXCEPTION 'SECURITY VIOLATION: Only a verified Director may alter institutional roles.';
  END IF;
  RETURN NEW;
END;
$$ LANGUAGE plpgsql SECURITY DEFINER;
```

POSTGRESQL 15 + ROW-LEVEL SECURITY

Fine-Grained Multi-Role Isolation at the Database Kernel

Even if a user manipulates client-side JavaScript or crafts raw REST API requests, PostgreSQL evaluates the caller's signed JWT claims against strict Row-Level Security (RLS) policies before returning or mutating a single row.

ROW-LEVEL SECURITY (RLS) ACCESS CONTROL MATRIX ACROSS CORE TABLES

POSTGRESQL TABLE	STUDENT POLICY (`ROLE = 'STUDENT'`)	TEACHER POLICY (`ROLE = 'TEACHER'`)	DIRECTOR POLICY (`ROLE = 'DIRECTOR'`)
`profiles`	SELECT all; UPDATE own (role locked)	SELECT all; UPDATE own & cohort students	**Full SELECT / INSERT / UPDATE**
`attendance_records`	SELECT own (`student_id = auth.uid()`)	SELECT / INSERT / UPDATE assigned class	**Full Institutional Access**
`grades`	SELECT own (`student_id = auth.uid()`)	SELECT / INSERT / UPDATE assigned class	**Full Institutional Access**
`assignments` & `submissions`	SELECT class work; UPDATE own submission	Full CRUD for taught cohorts	**Full Institutional Access**
`ventures` & `venture_votes`	INSERT own pitch; 1 vote per venture	SELECT all student pitches	**Full Oversight**
`leave_records` & `substitutions`	No Access (0 rows returned)	INSERT / SELECT own leave requests	**Approve / Reject & Assign Subs**
`person_documents`	SELECT own verified certificates	SELECT / UPLOAD for assigned cohort	**Full Dossier & Contract Control**
`audit_logs`	INSERT own action events only	INSERT own action events only	**Full Forensic SELECT (Append-Only)**

MIGRATION 01–04

Core Schema & Realtime

Initial tables, enums, indexes, auth triggers, RLS policies, and Supabase Realtime publications.

MIGRATION 05–07

Leave, Planner & Buckets

Faculty leave workflows, lecture planner tables, grouped period attendance, and private document storage buckets.

AUTOMATED AUDIT

`scripts/check-rls.js`

Automated verification script asserting `ENABLE ROW LEVEL SECURITY` across all 24 tables prior to release.

PLUG-AND-PLAY CURRICULUM TOPOLOGY

1-Click Board Presets: CBSE, ICSE, IB, Cambridge IGCSE & State Boards

Unlike rigid ERPs that hardcode "Class XI-B" or fixed subjects into source code, our **Academic Setup Engine (`academic-setup.js`)** makes the entire educational structure 100% database- and config-driven.

8 CONFIGURABLE ACADEMIC ENTITIES (FULL CRUD)

1. Grade Levels	e.g. Class I–XII, Grade 1–12, DP1–DP2, Year 7–13
2. Sections & Divisions	Capacity-tracked cohorts (A, B, Set 1, Alpha)
3. Academic Streams	Science, Commerce, Humanities, IB Groups 1–6
4. Subject Catalog	Core, Elective, Language & Co-Curricular codes
5. Evaluation Terms	Semesters, Quarters, Term 1 & 2 with weightage %
6. Assessment Types	Unit Tests, Internal Assessment, Practicals, Finals
7. Grading Scales	CBSE A1–E, IB 1–7, IGCSE A*–G, Percentage
8. House System	Custom Houses with live point ledgers & colors

BUILT-IN 1-CLICK INSTITUTIONAL BOARD PRESETS

CBSE (National Curriculum, India)

Classes I–XII • Science/Commerce/Humanities • UT1, Half-Yearly, UT2, Annual (A1–E Nine-Point Scale)

ICSE / ISC (CISCE Board)

Classes I–XII • Internal Assessments + Semester Exams • 1–9 Point Credit Scale

IB (International Baccalaureate MYP & DP)

MYP 1–5 & DP 1–2 • 6 Subject Groups + TOK/EE/CAS • Criterion A–D & 1–7 Scale

Cambridge Assessment (IGCSE & A-Levels)

Years 7–13 • Core & Extended Tracks • Mock, Coursework & Series Exams (A*–U)

FIRST-RUN ACADEMIC SETUP WIZARD & PORTABLE JSON SNAPSHOTS

WIZARD STEP 1

Select Board Preset

Choose CBSE, ICSE, IB, Cambridge, State Board, or Blank Custom Canvas.

→

WIZARD STEP 2

Tailor Cohorts & Houses

Customize sections, capacities, streams, subjects, and house crests.

→

WIZARD STEP 3

Toggle Feature Flags

Enable/disable Pitch Pad, House Cup, Leave Engine, or Broadcasts.

→

WIZARD STEP 4

Export / Commit

Commits to PostgreSQL & exports portable JSON school blueprint.

STUDENT-CENTRIC DESIGN

A Calm, High-Clarity Academic Cockpit for Every Scholar

When a student signs in, the operating system renders a personalized command center surfacing their daily schedule, real-time attendance standing, continuous assessment trajectory, coursework deadlines, and verified digital identity pass.



CORE SCHOLAR CAPABILITIES

Personal Attendance Ledger	Subject-wise & daily attendance % with threshold alerts
Interactive Timetable	Live period highlighting & room/faculty mapping
Coursework & Submissions	Track due dates, download attachments & mark completion
Assessment Analytics	View Unit Test, Half-Yearly & Final Exam breakdowns
House Cup Standings	Live points leaderboard across campus houses
Verified Document Vault	Access official report cards & certificates

PRIVACY BY DESIGN: ZERO HORIZONTAL PEER LEAKAGE

Unlike shared spreadsheets or poorly secured portals where students can guess URLs to view classmates' grades, every student query is filtered by PostgreSQL RLS policy `USING (auth.uid() = student_id)`. A scholar can inspect their own longitudinal growth in rich detail while having **zero visibility** into any peer's attendance, marks, or personal dossier.

BEYOND GRADES & ATTENDANCE

The Pitch Pad: A Built-In Campus Startup & Project Incubator

Schools should not merely grade exams — they should incubate builders. **The Pitch Pad** enables scholars to propose campus ventures, recruit co-founders, and gather democratic peer backing verified by cryptographic database triggers.

FEATURED STUDENT VENTURES IN ECOSYSTEM

CampusCart & QueueLess
Founder: Ishaan Kalra • Smart Canteen Pre-Ordering
▲ 48 VOTES

SkillSync Peer Tutoring
Founder: Ananya Gupta • Cross-Grade Study Exchange
▲ 42 VOTES

EcoTrack Campus Carbon
Founder: Diya Sen • Solar & Waste Telemetry
▲ 39 VOTES

StudySphere & ClubFlow
Founders: Kabir Sharma & Rohan Varma
▲ 35 VOTES

ANTI-TAMPERING VOTE INTEGRITY ENGINE

To prevent technical students from spoofing vote counts via browser DevTools, direct updates to `ventures.votes` are blocked by a PostgreSQL trigger. Votes are strictly computed from the relational `venture_votes` table with a unique composite key `UNIQUE(venture_id, student_id)`.

```
-- Trigger: Recomputes exact vote count on INSERT/DELETE
CREATE TRIGGER tr_sync_venture_votes
AFTER INSERT OR DELETE ON public.venture_votes
FOR EACH ROW EXECUTE FUNCTION
public.sync_venture_votes();

-- Trigger: Blocks manual client tampering of vote column
CREATE TRIGGER tr_prevent_vote_tampering
BEFORE UPDATE ON public.ventures
FOR EACH ROW EXECUTE FUNCTION
public.prevent_manual_venture_vote_tampering();
```

> *"By embedding a peer-backed venture incubator directly inside the daily academic workspace, Campusly transforms passive students into active campus problem-solvers."*

DESIGNED FOR CLASSROOM VELOCITY

13 Purpose-Built Faculty Modules That Reclaim Teaching Time

Educators get an ergonomic, keyboard-navigable workspace (`Cmd/Ctrl + K` Command Palette included) that reduces daily attendance, lesson planning, and cohort administration from 45 minutes to under 90 seconds.

01 // ATTENDANCE

Period & Cohort Roll-Call

Mark Present, Absent, or Late with 1-click bulk actions, grouped session logs, and instant attendance report exports.

02 // ROSTER & RECORDS

Class Record Register

360° student profiles with guardian contacts, house affiliation, academic stream, and secure PDF document attachments.

03 // ASSIGNMENTS

Coursework & Attachments

Publish assignments with file uploads in storage buckets, due dates, and real-time student completion tracking.

04 // SYLLABUS PACING

Lecture & Test Planner

Map unit-wise lecture schedules, track syllabus completion %, and coordinate test dates across faculty.

05 // COMMUNICATION

Targeted Broadcasts

Dispatch high-priority announcements to specific cohorts or the entire campus with instant realtime push.

06 // STAFF MOBILITY

Faculty Leave Portal

Submit leave requests with date ranges and substitution notes directly to the Director's approval queue.

COHORT-SCOPED AUTHORIZATION (`PUBLIC.IS_TEACHER_OF()`)

Teachers are granted write privileges strictly for their assigned cohorts and subjects. Our custom SQL security function `public.is_teacher_of(student_class)` cross-references the authenticated faculty profile before permitting attendance or grade mutations, preventing accidental cross-class edits.

```
CREATE POLICY "teachers_manage_assigned_attendance"
ON public.attendance_records FOR ALL
USING (public.is_teacher_of(class_section) OR public.is_director());
```

REAL-TIME ACADEMIC TELEMETRY

Continuous Assessment Gradebook & Final Exam Tabulation

Supporting both formative continuous assessments (`page-teacher-gradebook`) and summative board/final examinations (`page-teacher-final-marks`), the Gradebook Engine eliminates manual formula errors and spreadsheet corruption.

LIVE GRADEBOOK ARCHITECTURE: MULTI-TERM WEIGHTED COMPUTATION

ASSESSMENT CYCLE	MAX MARKS / SCALE	TERM WEIGHTAGE	AUTOMATED COMPUTATIONS PERFORMED BY OS
Unit Test 1 & 2 (Formative)	20 / 25 / 50 Marks	10% – 20%	Instant percentage normalization, class average & cohort percentile rank
Half-Yearly / Mid-Term Exam	80 / 100 Marks	30% – 40%	Subject-wise dispersion analysis, grade band mapping (A1–E / 1–7)
Internal / Practical / Project	20 Marks	10% – 20%	Rubric validation, attachment verification & moderation audit log
Final Annual Examination	80 / 100 Marks	40% – 50%	**Cumulative GPA rollup & feeds Year-End Promotion Engine**

SAFEGUARDS AGAINST GRADE TAMPERING

- **Server-Validated Bounds:** Marks exceeding `max_score` or negative inputs are rejected at both form and database constraint levels.
- **Double-Submit Protection:** Idempotent async locks prevent duplicate grade insertions on rapid clicks.
- **Audit Trail on Every Edit:** Any modification to a published mark logs the actor ID, previous value, new value, and timestamp.

INSTANT EARLY-WARNING IDENTIFICATION

By correlating Period Attendance with Unit Test trajectories in real time, the system flags at-risk scholars before final exams — enabling proactive faculty mentorship rather than post-mortem failure reports.

100%
FORMULA ACCURACY

0.0s
REPORT CARD LAG

MACRO INSTITUTIONAL INTELLIGENCE

The Executive Board: Complete School Visibility in One Screen

Built for Principals, Directors, and Trustees, the Executive Board (`page-director-analytics`) synthesizes thousands of daily classroom interactions into actionable institutional telemetry.

14 EXECUTIVE DIRECTOR MODULES

01. Executive Analytics Board	Real-time campus attendance, GPA & cohort KPIs
02. Class Record Register	Master searchable directory of all scholars
03. Account Approvals Queue	Verify & activate pending user registrations
04. Faculty Roster & Dossiers	360° teacher profiles, qualifications & contracts
05. Faculty Provisioning	Invite educators with forced first-login reset
06. Bulk CSV Student Import	Batch provision hundreds of students in 1 click
07. Attendance Rollup Reports	Cross-grade attendance heatmaps & CSV export

GOVERNANCE, CURRICULUM & COMPLIANCE MODULES

08. Master Test Schedule	School-wide examination calendar & room matrix
09. Staff Leave & Substitutions	Approve leave & assign relief teachers
10. Curriculum & Syllabus Rollup	Monitor lecture pacing across all departments
11. School-Wide Feed	Executive broadcasts & institutional notices
12. Immutable Audit Logs	Forensic security ledger with actor filtering
13. Year-End Student Promotion	Automated cohort progression & overrides
14. Live Branding & Academic Setup	White-label identity & multi-board configuration

SECURE BATCH PROVISIONING: `PUBLIC.BULK_IMPORT_STUDENTS(JSONB)`

How do you onboard 2,000 students without exposing the Supabase `service_role` key in the browser? We engineered a PostgreSQL `SECURITY DEFINER` RPC function guarded by `IF NOT public.is_director() THEN RAISE EXCEPTION`. The Director uploads `students-import-template.csv`, and the database atomically provisions auth accounts, profiles, cohorts, and houses.

SOLVING REAL SCHOOL HEADACHES

Automated Morning Substitutions & Year-End Cohort Promotion

Two of the most chaotic moments in any school are **7:30 AM when a teacher calls in sick** and **March/April during year-end student promotions**. We engineered dedicated engines for both.

ENGINE A // STAFF LEAVE & SUBSTITUTION MATRIX

- **Structured Leave Filing:** Educators submit leave type (Medical, Casual, Academic Duty), date range, and handover notes.
- **1-Click Director Adjudication:** Principal reviews pending requests on `page-director-leave` with live conflict detection.
- **Smart Substitution Assignment:** Upon approval, the Director assigns a relief teacher for affected periods, updating the daily schedule immediately.
- **Complete Leave Ledger:** Tracks cumulative leave balances per faculty member across the academic year.

ENGINE B // YEAR-END STUDENT AUTO-PROMOTION

- **Rule-Driven Progression:** Evaluates cumulative final marks and attendance thresholds across every cohort on `page-director-promotion`.
- **Automated Cohort Mapping:** Maps Class XI-B → Class XII-B, or marks graduating seniors (Class XII → Alumni Archive).
- **Granular Principal Override:** Allows the Director to review edge cases, grant conditional promotion, or retain specific scholars before committing.
- **Historical Transcript Preservation:** Prior year grades and attendance remain permanently queryable.

360° FACULTY DOSSIERS & ENCRYPTED DOCUMENT VAULT

Clicking any educator in the Director's Faculty Roster opens `page-director-teacher-detail` — a comprehensive dossier displaying assigned classes, syllabus completion velocity, attendance marking consistency, leave history, and encrypted local storage attachments (employment contracts, degree certificates, background verifications).

TRUST THROUGH VERIFICATION

Every Administrative & Academic Mutation Is Permanently Logged

In an educational institution, grade integrity and roster changes must be above suspicion. Our **Immutable Audit Ledger (`page-director-audit`)** records every sensitive action into an append-only table where neither students, teachers, nor administrators have `DELETE` or `UPDATE` privileges.

LIVE AUDIT STREAM SAMPLE (APPEND-ONLY FORENSIC TELEMETRY)

UTC TIMESTAMP	ACTOR IDENTITY & ROLE	EVENT CATEGORY	TARGET RESOURCE & CRYPTOGRAPHIC SUMMARY
`2026-10-13 08:12:04`	`director@brainz.edu` (Director)	`BRANDING_UPDATE`	Updated institutional config & verified WCAG AA contrast
`2026-10-13 07:45:19`	`director@brainz.edu` (Director)	`BULK_IMPORT`	Executed `bulk_import_students()` — 120 scholars provisioned
`2026-10-13 07:30:02`	`teacher@brainz.edu` (Teacher)	`GRADE_COMMIT`	Committed Unit Test 2 marks for Cohort XI-B Economics
`2026-10-13 07:15:44`	`teacher@brainz.edu` (Teacher)	`ATTENDANCE_SYNC`	Marked Period 1 Attendance for XI-B (38 Present, 1 Absent)
`2026-10-13 06:58:10`	`director@brainz.edu` (Director)	`LEAVE_APPROVED`	Approved leave #LR-204 & assigned Period 3 substitution

APPEND-ONLY RLS POLICY ENFORCEMENT

```
-- Anyone authenticated can log their own action
CREATE POLICY "audit_insert_own" ON public.audit_logs
FOR INSERT WITH CHECK (auth.uid() = actor_id);

-- Only Director can read; NO UPDATE or DELETE policies exist!
CREATE POLICY "audit_select_director" ON public.audit_logs
FOR SELECT USING (public.is_director());
```

WHY JUDGES & AUDITORS LOVE THIS

Because PostgreSQL denies all `UPDATE` and `DELETE` operations on `public.audit_logs` by default when no policy is defined, the audit trail is mathematically append-only at the application layer — satisfying both board accreditation inspections and DPDP Act Section 8 forensic readiness.

ARCHITECTURAL EDITORIAL PALETTE (WW/STUDIO)

A Museum-Grade Design System With Real-Time White-Labeling

Enterprise school software doesn't have to look like a 1998 gray spreadsheet. Our editorial monograph & live branding engine (`page-director-branding`) harmonizes tactile warm paper tones (**White**, **Almond**, **Azure Mist**) with grounded architectural accents (**Terracotta**, **Steel**, **Rosemary**).

#FFFFFF	#F2EDE7	#E8EBED	#B78E79	#414751	#919F90
White	**Almond**	**Azure Mist**	**Terracotta**	**Steel**	**Rosemary**
Elevated Card Surface & Crisp Panel Canvas	Warm Editorial Paper Foundation (Zero Glare)	Secondary Container Fill & Structural Grid	Primary Architectural Accent, Focus & Callouts	Deep Structural Ink & High-Contrast Typography	Botanical Sage Status & Verification Accent

LIVE CONFIG-DRIVEN BRANDING ENGINE

- **Instant White-Labeling:** The Director can customize Product Name, School Name, Tagline, Crest Monogram/Logo, and Accent Palette in real time.
- **Built-In WCAG Contrast Calculator:** Automatically computes relative luminance ratios as colors are tweaked and displays a live `WCAG AA Compliant` shield.
- **Zero-Reload Propagation:** Updates DOM titles, OpenGraph metadata, sidebar wordmarks, and print report headers immediately.

ERGONOMICS & ACCESSIBILITY EXCELLENCE

- **Command Palette (`Cmd/Ctrl + K`):** Instant fuzzy-search navigation across all role-permitted views and actions.
- **100% Self-Hosted Fonts & Icons:** All 28 `.woff2` font weights and Lucide vector icons are vendored locally in `/vendor/` for zero FOUT and offline speed.
- **ARIA Landmarks & Keyboard Focus:** Explicit `aria-label` attributes, modal focus handling, and `prefers-reduced-motion` support.

BUILT FOR STATUTORY SOVEREIGNTY

Full Compliance with India's Digital Personal Data Protection (DPDP) Act, 2023

Under the DPDP Act 2023, every person under 18 is classified as a protected child Data Principal, imposing severe financial penalties (up to ₹250 Crore) on institutions that leak minor data or perform unauthorized behavioural tracking.

STATUTORY COMPLIANCE MAPPING MATRIX

DPDP ACT 2023 PROVISION	STATUTORY MANDATE	ARCHITECTURAL IMPLEMENTATION IN CAMPUSLY / EDUVIA
Section 9(1): Children's Data	Verifiable parental/guardian consent prior to processing any child's personal data.	**Schema enforces `parent_consent_verified` timestamp & guardian contact verification before account activation.**
Section 9(3): No Tracking of Minors	Strict prohibition on behavioural monitoring or targeted advertising to children.	**Zero third-party analytics, zero ad pixels, zero external cookies. 100% air-gapped operational telemetry only.**
Section 6 & 8: Purpose Limitation	Collect only minimum data required for stated educational purpose; protect against breaches.	**Stores strictly academic records (attendance, grades, timetable). Protected by PostgreSQL RLS & HS256 JWTs.**
Section 12: Right to Correction & Erasure	Parents/Data Principals have the right to correct inaccurate data or erase records upon withdrawal.	**Director console supports instant profile correction, graduation archival, and permanent dossier erasure.**
Section 16: Cross-Border Data Residency	Restrictions on transferring sensitive personal data outside sovereign boundaries.	**100% of database & file volumes reside physically inside the school's own server room in India.**

> *"While cloud SaaS competitors struggle to retrofit legal compliance onto shared foreign databases, our air-gapped on-premise architecture makes DPDP Act 2023 compliance native by physics."*

ZERO SINGLE POINTS OF FAILURE

Cryptographically Verified Backups & 100% Offline Asset Vendoring

Running on-premise requires enterprise-grade disaster recovery. We built automated cross-platform backup/restore pipelines (`backup-db.sh` / `backup-db.ps1`) and complete local CDN asset vendoring (`scripts/vendor-assets.js`).

AUTOMATED 02:00 AM BACKUP & RESTORE PIPELINE

Archive Format	PostgreSQL Custom Compressed (`pg_dump -Fc`)
Cryptographic Checksum	Companion `.sha256` hash generated per dump
Rolling Retention Policy	Automated 14-day snapshot retention & pruning
Pre-Restore Verification	`restore-db.ps1` validates SHA-256 before restore
Clean Production Wipe	`supabase/reset-to-clean.sql` truncates demo seeds

```
# Execute Verified Backup & SHA-256 Checksum
PS> .scripts□ackup-db.ps1
[OK] Saved: ./backups/eduvia_backup_20261013_020000.dump
[OK] SHA256: 9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b...
```

100% LOCAL ASSET VENDORING (ZERO CDN DEPENDENCY)

Most web apps break offline because they fetch Google Fonts, Unpkg icons, or Supabase client scripts from the internet. We vendored every byte locally:

vendor/fonts/ (28 files)	All Newsreader & Plus Jakarta Sans `.woff2`
vendor/fonts.css	Local `@font-face` rules (0 calls to googleapis)
vendor/lucide.min.js	Complete vector icon library served from disk
vendor/supabase.min.js	Official Supabase JS client vendored locally

0
EXTERNAL DNS LOOKUPS

100%
AIR-GAP UPTIME

PROVEN BY AUTOMATED TESTS

End-to-End Playwright Verification & Static Codebase Audits

We did not stop at building features — we engineered an extensive suite of **Playwright browser automation tests** and **40+ Node.js verification scripts** to mathematically prove zero dead controls, zero external network leaks, and 100% role isolation.

AUTOMATED VERIFICATION & TEST SUITES

e2e-no-dead-controls.spec.js	Asserts every button/link has a valid handler & 0 external calls
director-tabs.spec.js	Walks all 14 Director views verifying data hydration
scripts/check-rls.js	Parses SQL migrations to assert 100% RLS table coverage
scripts/scan-xss.js	Audits DOM sinks to ensure strict HTML escaping
scripts/secret-scan.js	Scans entire repository for high-entropy credentials
scripts/check-dom-refs.js	Verifies every `getElementById` target exists in DOM

AUDIT REMEDIATION SCORECARD (PRE-PROD VS. FINAL)

AUDIT METRIC	INITIAL PROTOTYPE	PRODUCTION V2.4
External CDN Requests	14 Requests	**0 Requests (100% Local)**
Hardcoded Class/Board Strings	50+ Occurrences	**0 (Dynamic Academic Setup)**
Unescaped XSS Sinks	Multiple	**0 (Sanitized via `escapeHtml`)**
Director Navigation Race Condition	Present	**Resolved & E2E Locked**
RLS Enabled Tables	Partial	**24 / 24 Tables (100%)**

30+
VERIFIED OS VIEWS

40+
AUTOMATED AUDIT SCRIPTS

0
DEAD UI CONTROLS

100%
ROLE GUARD PASS RATE

THE ECONOMIC & EDUCATIONAL CASE

85% Lower 5-Year Total Cost of Ownership + Complete Data Sovereignty

By eliminating recurring per-student cloud SaaS subscriptions and running on existing school server hardware, **Campusly / Eduvia** saves a mid-sized school (1,500 students) over **₹60 Lakhs ($75,000+)** over 5 years while guaranteeing absolute privacy.

5-YEAR TCO COMPARISON (1,500-STUDENT SCHOOL)

COST CATEGORY	LEGACY CLOUD ERP	CAMPUSLY / EDUVIA
Annual SaaS Licensing	₹12,00,000 / yr	**₹0 (Self-Hosted Open Core)**
Per-Module Add-On Fees	₹3,50,000 / yr	**₹0 (All 30+ Views Included)**
Data Export / Exit Fee	Vendor Locked	**₹0 (Full SQL & CSV Ownership)**
5-Year Cumulative Spend	**₹77,50,000+**	**Local Server Hardware Only**

NEXT-HORIZON R&D ROADMAP (V3.0 SYNAPSE)

Phase 1 (Current v2.4)	Air-Gapped 3-Role OS, 24-Table RLS, Multi-Board Engine
Phase 2 (Q1 2027)	Local Ollama / pgvector AI Tutor running 100% on school GPU
Phase 3 (Q2 2027)	Offline NFC / Smart-Card Gate Attendance Hardware Bridge
Phase 4 (Q4 2027)	Federated Inter-School Olympiad & Pitch Pad Network

THANK YOU TO THE JUDGING PANEL

CAMPUSLY // EDUVIA — THE SOVEREIGN SCHOOL OS

VERIFIED BUILD v2.4 2026

Live Interactive Demo Available Immediately • Student, Teacher & Director Portals Ready for Inspection